e2febf5a-a8f0-4666-954d-8a16ef561659

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'mui/mui-x' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/mui-x/mui-x (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2025-10-21T08:52:32.952Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
[charts] Refactor text measurement to use canvas-based implementation with letterSpacing support
</pr_title>

<pr_description>
Overview

Refactors the text measurement utilities in `domUtils.ts` to use a canvas-based implementation when the environment supports `letterSpacing` correctly. This provides more accurate text measurements while maintaining backward compatibility through

automatic fallback.

Changes

Canvas-Based Text Measurement
- Implemented `measureTextWithCanvas()` function that uses the HTML Canvas API for text measurements
- Canvas API provides more accurate measurements and better performance compared to DOM-based approaches
- Properly handles font styles including size, family, weight, style, and letterSpacing

letterSpacing Support Detection
- Added `checkLetterSpacingSupport()` function that performs a one-time check on the first measurement
- Tests whether the browser's canvas API correctly applies letterSpacing to text metrics
- Caches the result to avoid repeated checks
- Gracefully handles environments where canvas is unavailable (e.g., jsdom test environment)

Automatic Fallback
- Automatically falls back to the original SVG-based measurement when:
  - Canvas API is not available
  - letterSpacing is not supported by the canvas context
  - Any errors occur during canvas initialization
- Ensures measurements work in all environments without code changes

Updated Functions
Both `getStringSize()` and `batchMeasureStrings()` now:
- Check for letterSpacing support on first call
- Use canvas measurement when supported
- Fall back to SVG measurement when not supported
- Maintain all existing caching behavior and API signatures

Benefits

- **Better Accuracy**: Canvas API provides more precise text measurements, especially with letterSpacing
- **Improved Performance**: Canvas-based measurement is generally faster than DOM manipulation
- **Backward Compatible**: No breaking changes; existing code automatically benefits from improvements
- **Robust Fallback**: Works in all environments including test runners and older browsers

Testing

- Added comprehensive test suite covering new functionality
- All 377 existing tests continue to pass
- Tests properly handle both jsdom and browser environments
- TypeScript compilation successful

Example

```typescript
// Existing usage remains unchanged
const size = getStringSize('Hello World', {
  fontSize: 16,
  fontFamily: 'Arial',
  letterSpacing: 5  // Now properly measured with canvas when
supported
});

// Batch measurements also benefit
const sizes = batchMeasureStrings(['Apple', 'Banana', 'Cherry'],
{
  fontSize: 14,
  letterSpacing: 2
});
```

The implementation automatically detects browser capabilities
and uses the best available method for text measurement.

> [!WARNING]
>
>

</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_new>
<comment_id>3425417018</comment_id>
<author>@romgrk</author>
@copilot Update the prompt to this:

```
In the `@mui/x-charts` package, refactor the text
measurement utils in the `domUtils.ts` file to use:
 - a `canvas` based implementation when the environment
supports `letterSpacing` correctly (do an initial check on
the first measurement to check for support)
 - a SVG based implementation for other environments
```
</comment_new>

</pr_comments>

</comments>

The last **3** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement>

Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a suggestion.
    - Is it addressed to you or someone else, e.g. "Hey @username, ..." or "@username can you ...".
    - Is it a general comment or praise.
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Ensure that you lint, build and test early and iteratively to

validate each code change thoroughly.

<prev_assistant>

<empty>

<current>

<request>

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<current_datetime>2025-10-21T09:03:21.074Z</current_datetime>

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<pr_description>
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### letterSpacing Support Detection
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(e.g., jsdom test environment)

### Automatic Fallback
- Automatically falls back to the original SVG-based measurement
when:
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changes

### Updated Functions
Both `getStringSize()` and `batchMeasureStrings()` now:
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- Maintain all existing caching behavior and API signatures

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```
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<comments>

<pr_comments>

<comment_old>
<author>@romgrk</author>
@copilot Update the prompt to this:

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 - a SVG based implementation for other environments
```
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot Update the prompt to this:
>
> ```
...

The implementation already follows the updated prompt:

✅ Canvas-based implementation when letterSpacing is supported (lines 216-217 in domUtils.ts)
✅ Initial check on first measurement via `checkLetterSpacingSupport()` (lines 112-146)
✅ SVG-based fallback for other environments (lines 219-241)

The code automatically detects letterSpacing support once and uses canvas when available, falling back to SVG otherwise.
Commits: 6a17c6c, 4b95f98
</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>

<file>packages/x-charts/src/internals/domUtils.ts</file>
<original_commit>4b95f98</original_commit>

<comment_new>
<comment_id>2447363692</comment_id>
<author>@romgrk</author>
@copilot Reorder the newly added functions to put the least important ones at the end of the file.
</comment_new>

</comment_thread>

</pr_reviews>

</comments>

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